BACHELOR'S THESIS · 2026

Speech / Music Classification

On-line three-class audio classification under streaming constraints

AUTHOR
Marek Hric

SUPERVISOR
Ing. Vladimír Malenovský, Ph.D.

INSTITUTE
Brno University of Technology · FIT

BRNO UNIVERSITY OF TECHNOLOGY · FACULTY OF INFORMATION TECHNOLOGY

01 / PROBLEM

Real-time speech vs. music vs. background.

Classify each audio frame as Speech, Music, or Background (silence and ambient noise) from a live stream, with no lookahead and cheap enough for a single CPU thread. The third class lets the classifier stand alone in front of a codec, handling silence and noise directly.

STREAMING CONSTRAINTS

- **Causal.** Past and present samples only.
- **Real-time.** Each step finishes within the data's own duration.
- **Low delay.** Per-decision hop 10–23 ms; first output ≤ 58 ms.

02 / APPROACH

Four references, then exploring the TCN design space.

REFERENCE MODELS · IMPLEMENTED FOR 3-CLASS STREAMING

DT
Lavner & Ruinskiy, 2009
Learned CART tree over 19 time/frequency features with ANOVA-F selection; exponential-decay smoothing on a 30-frame decision deque.

GMM
Khonglah & Prasanna, 2016
Three 8-component diagonal GMMs (one per class) over NAPS, PSR, log-mel, and modulation features; 1 s rolling log-likelihood window.

SVM
Khonglah & Prasanna, 2016
RBF kernel (C = 1, γ = 3), native 3-class via sklearn OvO; 20-frame rolling decision-vector buffer.

TCN
Lemaire & Holzapfel, 2019
Dilated 1-D causal convolutions with WeightNorm and residual blocks; 16 filters, 3 stacks × 4 blocks, kernel 5; RF ≈ 8.4 s.

PROPOSED · DESIGN-SPACE EXPLORATION OF THE CAUSAL TCN

TCN-L High-accuracy large variant
Δ^2 log-mel frontend + 1-D conv preprocessor + reference TCN backbone + LSTM head; 16 filters, 3 stacks × 4 blocks, kernel 5; RF ≈ 8.4 s; ~389 K parameters. Strongest score on the test split.

TCN-S Lightweight variant · primary contribution
Δ^2 log-mel frontend + reduced TCN backbone; 8 filters, 1 stack × 4 blocks, kernel 5; RF ≈ 2.8 s; ~4.6 K parameters, 1/7 the reference TCN at ~21× less per-frame compute, matching it within 0.003 macro F1.

03 / DATASET

100 h of frame-labeled audio, built from four public sources.

- **Deterministic build** from four HuggingFace sources.
- **Frame-level interval labels** with an 84 / 8 / 8 train / val / test split applied per source per subclass.
- **Synthetic augmentation** mixes in four synthetic speech subclasses at build time.

SOURCES

- **LibriSpeech:** Speech, clean and noisy.
- **FMA + Bel Canto:** Music, six genres + a cappella, equal mix.
- **DEMAND:** Background, environmental noise.

COMPOSITION · 6000 MIN (100 H) FULL TIER

Speech 40%	Music 40%	Background 20%
2400 min	2400 min	1200 min
clean 960 · dirty 480 · noisy 480 · multispeaker 180 · SoM 180 · MSoM 120	instrumental · electronic · pop · rock · hip-hop · folk · a cappella ≈ 343 min each	environmental noise (DEMAND)

04 / RESULTS

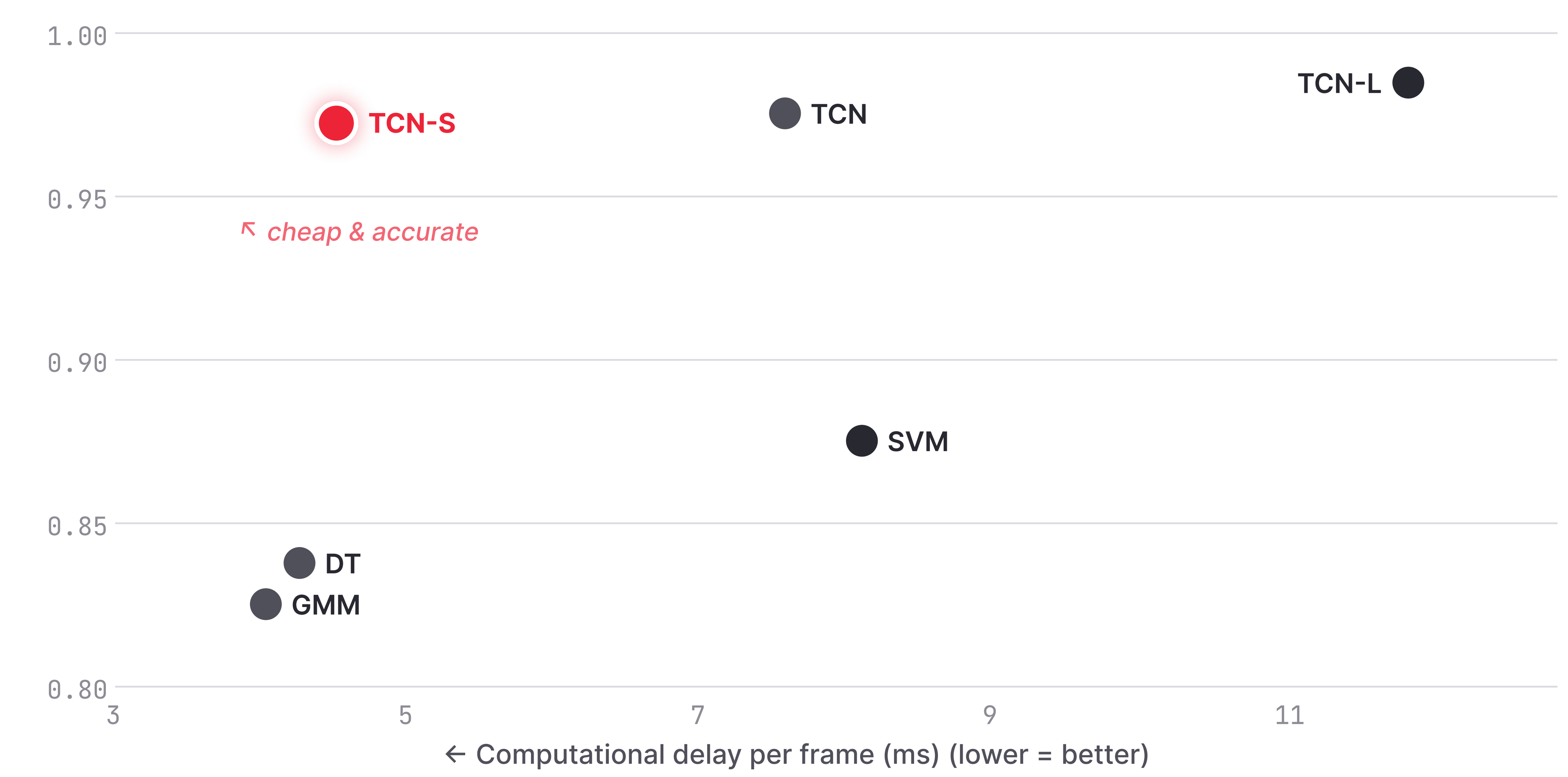
TCN-S matches the reference TCN at 1/7 the parameters.

TEST SPLIT · INTEL i5-8300H · 1 THREAD

MODEL	PARAMS	MACRO F1	RTF
TCN-L	389 K	0.9846	2.1×
TCN	33 K	0.9752	3.2×
TCN-S	4.6 K	0.9722	5.3×
SVM	521 K	0.8750	2.0×
DT	4.3 M	0.8376	2.4×
GMM	474	0.8250	3.8×

Every model clears 2× real-time. The TCN family adds 10–12 pp of macro F1 over the traditional references.

COST × EFFECTIVENESS



05 / CONTRIBUTIONS

A small model, a reproducible dataset, four streaming references.

PRIMARY
TCN-S: a 4.6 K-parameter streaming classifier.
Matches the reference TCN within 0.003 macro F1 at 1/7 the parameter count and ~21× less per-frame compute. Real-time factor 5.3× on a single thread of a 7-year-old laptop CPU.

SUPPORTING
Dataset building pipeline.
Deterministic multi-config dataset builder from HuggingFace sources with augmentation support, easily extensible by adding adapter downloaders for new sources.

SUPPORTING
Four 3-class on-line references.
Decision tree, GMM, and SVM sklearn implementations extended from 2-class to 3-class streaming, plus the causal TCN in PyTorch.